

Programming a Logi mouse with your device



Context

Why a mouse?

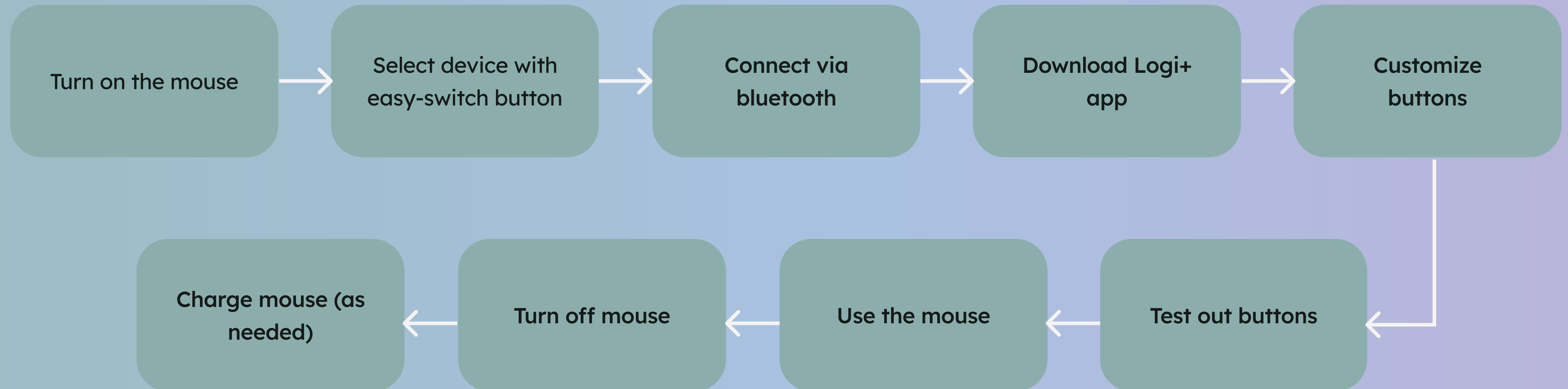
Wide range of accessible tech options available!



How does a specific mouse
exclude or include people in its
physical or digital interface?

*We will focus less on the physical mode since there are specific solutions available

High-level task breakdown

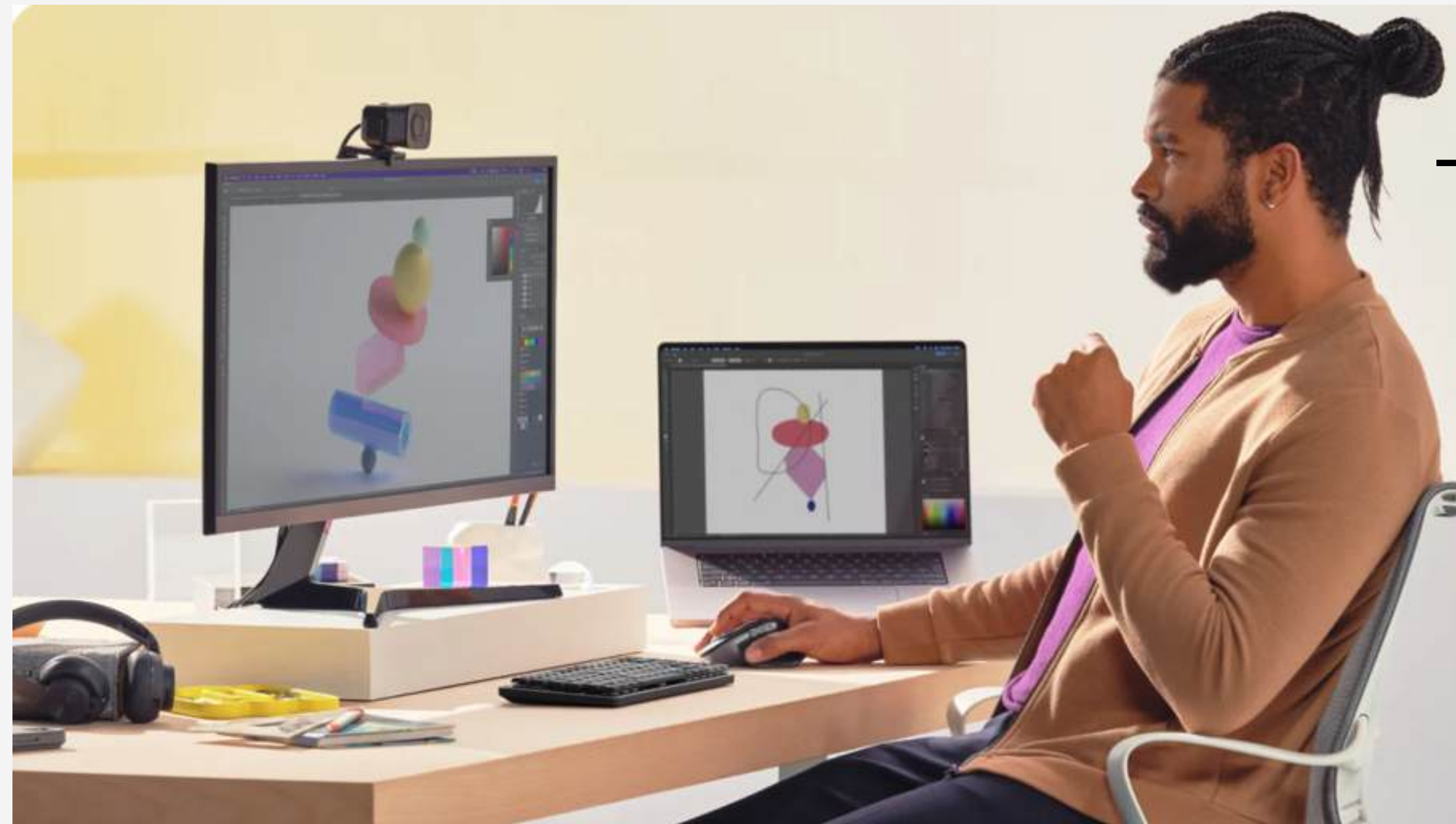


Key Findings

Key-Findings

Theme 1: A very narrow user is in mind

A specific audience was determined with little thought of the diversity within that group



At home, wealthy, able-bodied, non-colorblind, online, creative, young, professionals

What about the different modes?

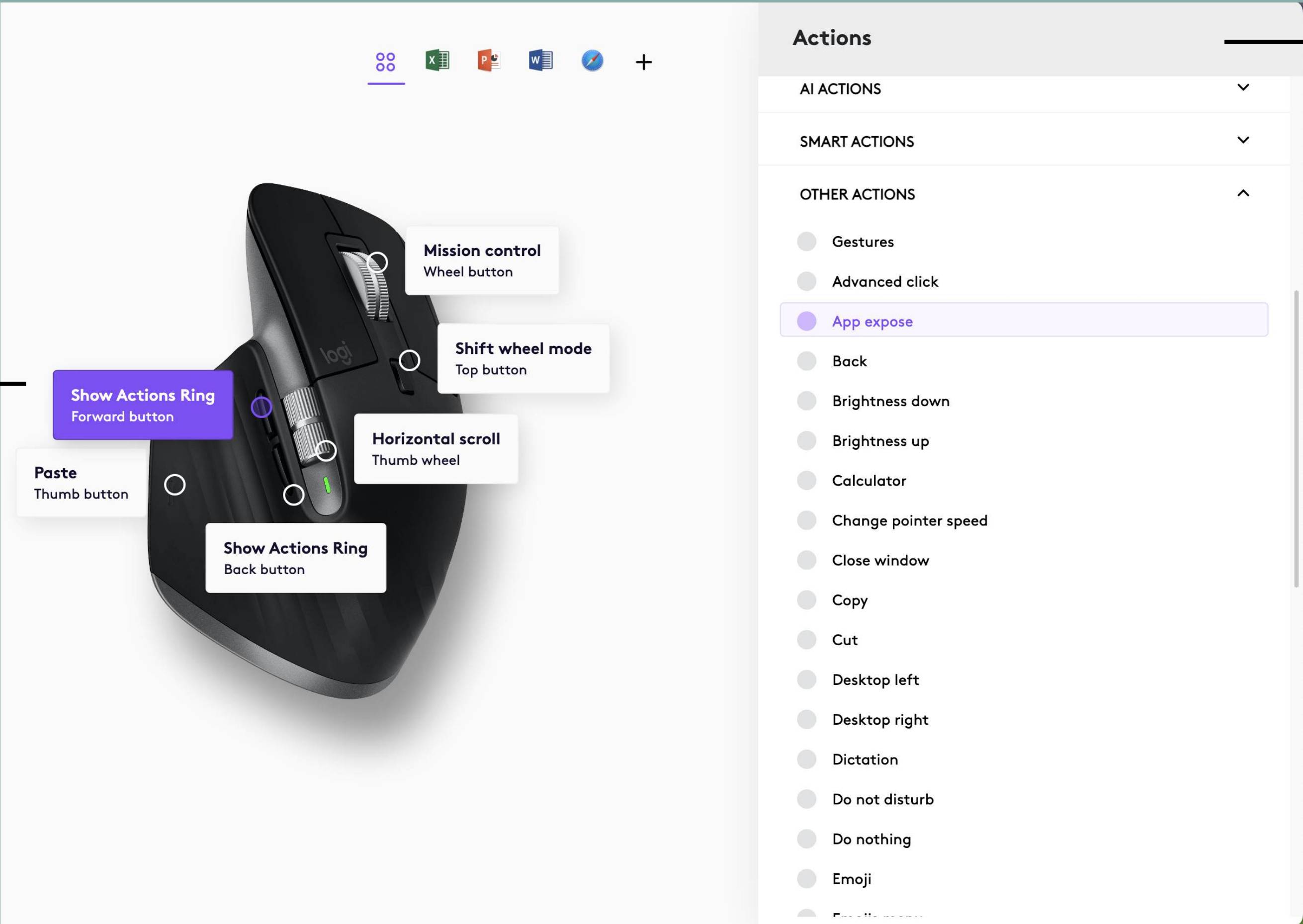
Key-Findings

Theme 2: The barrier to entry for learning is large

You must learn how to use the mouse **and** the app

programmable buttons

What does each one do?

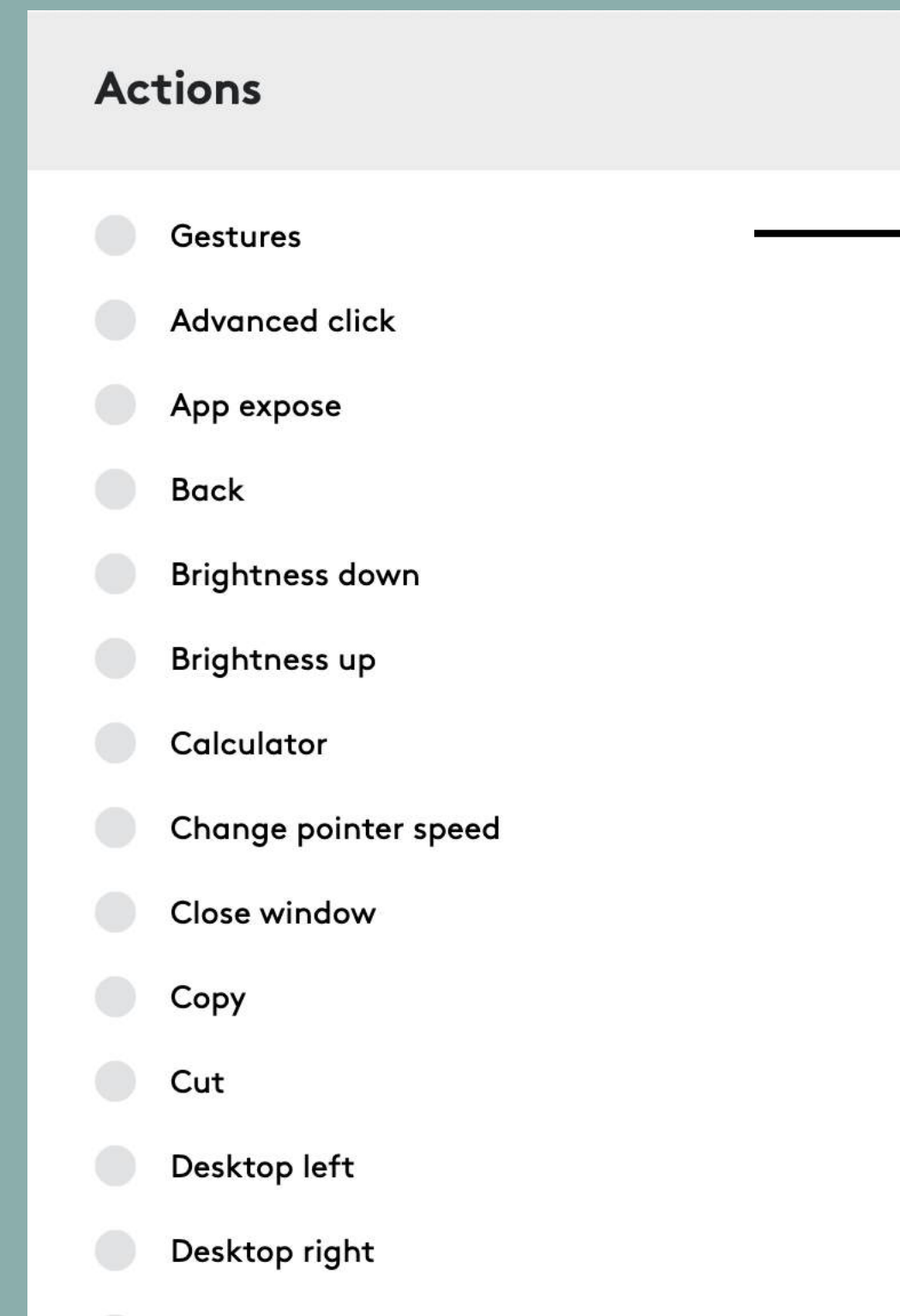


programmable actions

Key-Findings

Theme 2: The barrier to entry for learning is large

People are left to understand what each action means with no support



no pop-up or hover to give descriptions

Key-Findings

Theme 2: The barrier to entry for learning is large

Lack of support for helping people remember what each button does

Learning novel actions is HARD. It requires shifting them to the automatic functions of System One in our brain!

Johnson, J. (2014). Designing with the mind in mind: Simple guide to understanding user interface design guidelines (2nd ed.). Morgan Kaufmann.

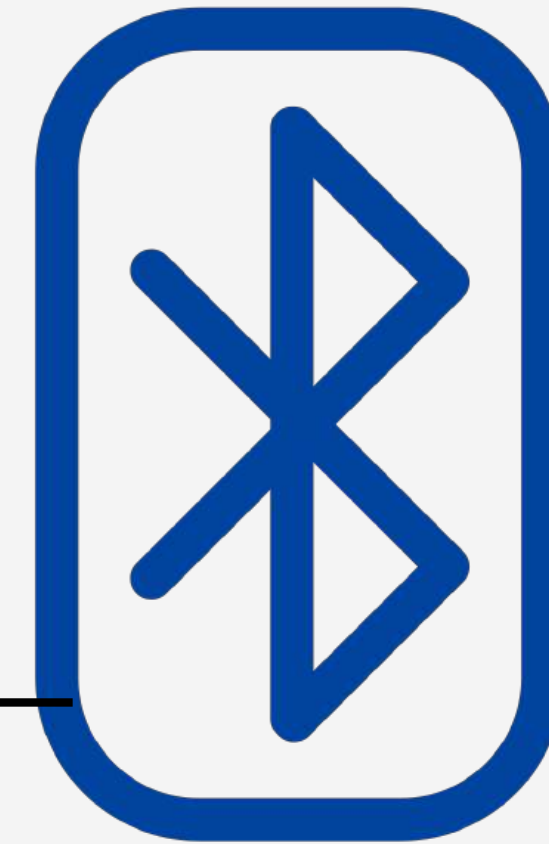
Key-Findings

Theme 3: The design is more exclusive than inclusive

There's an assumption that you must own a personal computer



requires Logi+ app to program buttons



bluetooth needed to connect to the mouse

Excludes:

- Those relying on public computer
(library, computer lab)

Key-Findings

Theme 3: The design is more exclusive than inclusive

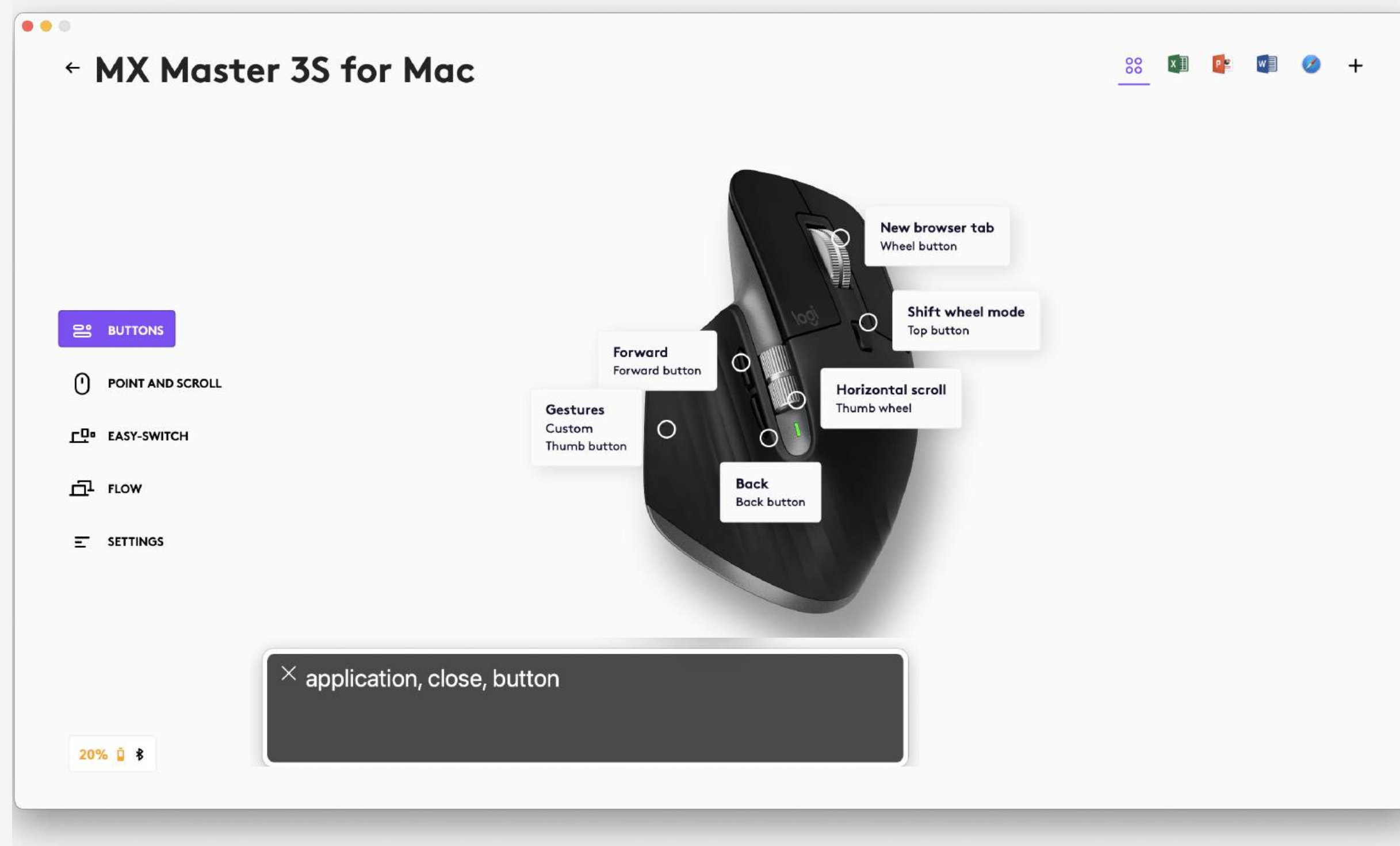
**Marketed as customizable, but limited to actions/
models Logitech has created**

No accessible options for those
using assistive technology (i.e,
start/stop voiceover, navigate
screen with buttons, enlarge text)

Key-Findings

Theme 3: The design is more exclusive than inclusive

Navigating the app with a screen reader is almost impossible



Excludes:

- Those with vision impairment

Key-Findings

Theme 3: The physical mouse does not support the experience

There's a lack of confirmation for completed actions



No feedback (auditory, visual, haptic):

- When switching between devices
- When turning on/off
- If left on after use!!

Key-Findings

Theme 3: The physical mouse does not support the experience

Unlabeled buttons provide no cognitive support



What button does what? What am I connected to?

What would a better experience require?

Inclusion Principle #1

Consider diversity that exists within an audience, even if you are narrow in some regards

Narrow audience?

Wealthy, able-bodied,
creative professional

Consider:

Cognitive/learning differences

Perceptual differences

Social context of use

Make buttons adaptable to each unique situation

What this looks like?

- Ability to program buttons beyond basic actions
- Support in-app to show how to program

Make learning easier by providing practice & reminders

What this looks like?

- Practice actions in-app
- Reminders for what buttons are programmed to do

We learn faster
when practice is
frequent, regular,
and precise!

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Inclusion Principle #4

Provide confirmation of actions to alleviate uncertainty

What this looks like?

- Haptic feedback when action completed
- Sound to confirm the mouse is connected

Moving to an improved experience

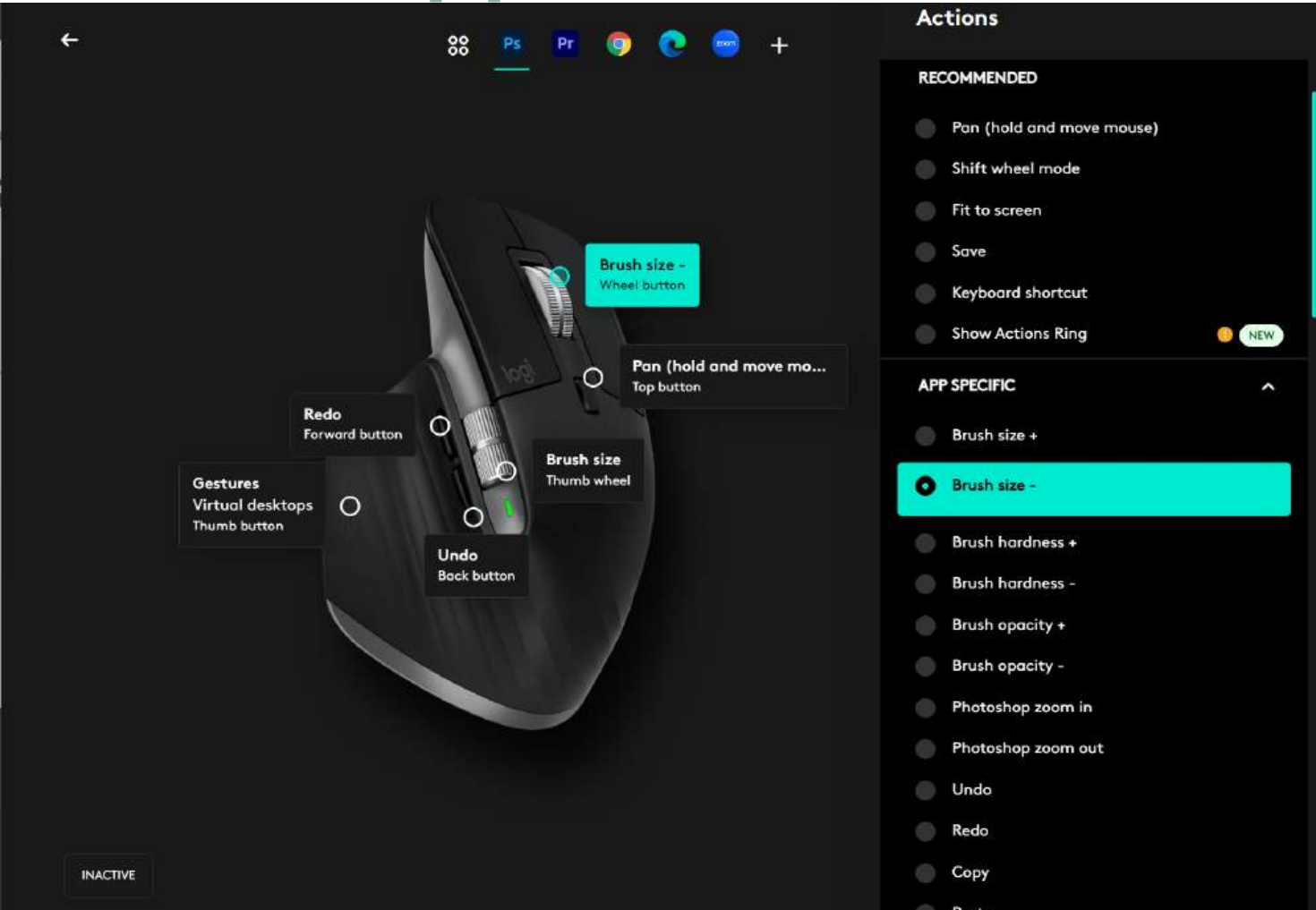
Concept #1: Digital Experience

Sandbox Mode

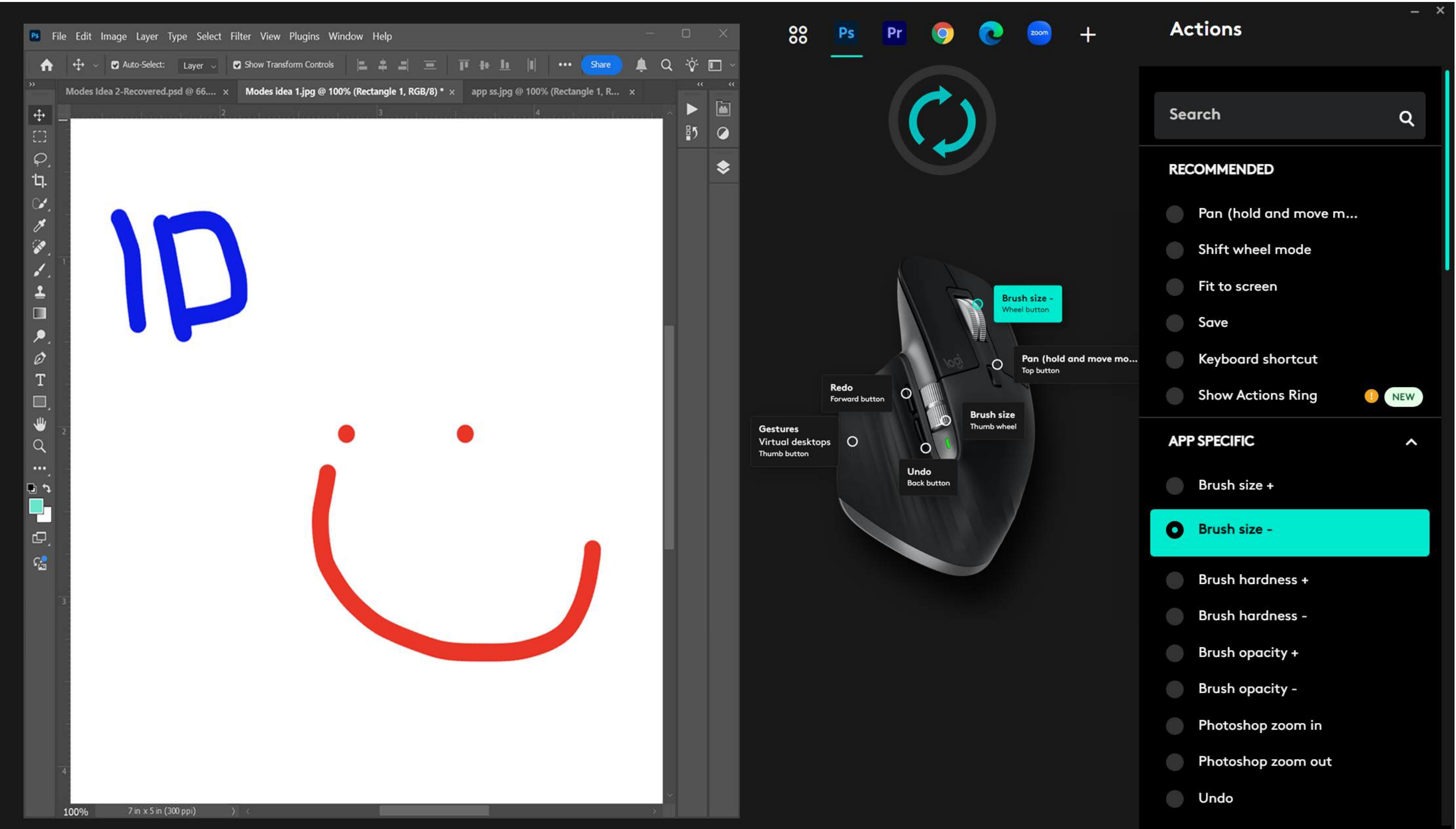
Inclusion Principle #2 & 3

- Reduces Need To Exit and Enter Apps to Test Buttons
- Helps with Memory Recall and Learning

Current App UI:



Final Proposition:



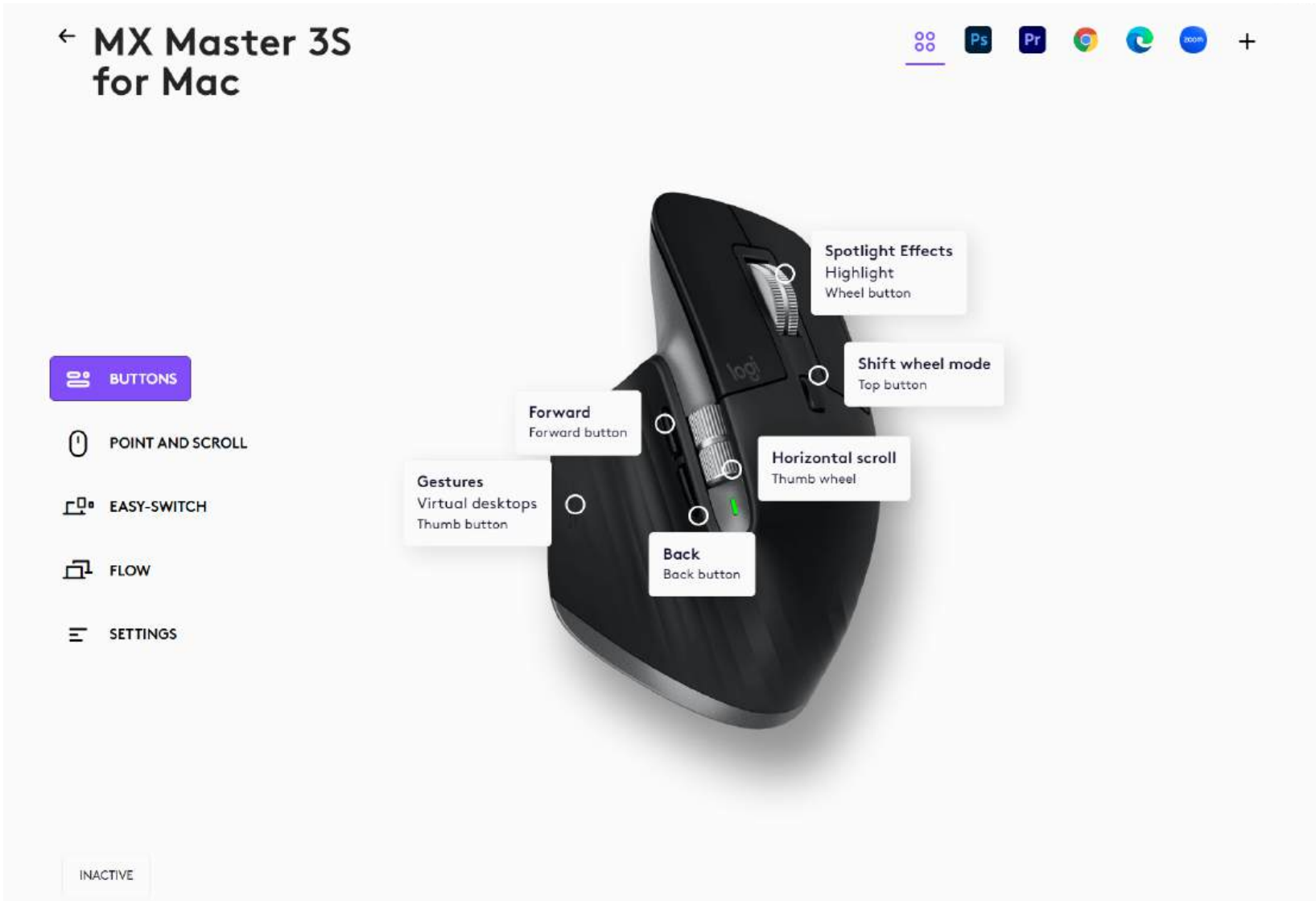
Concept #2: Digital Experience

Mini-map

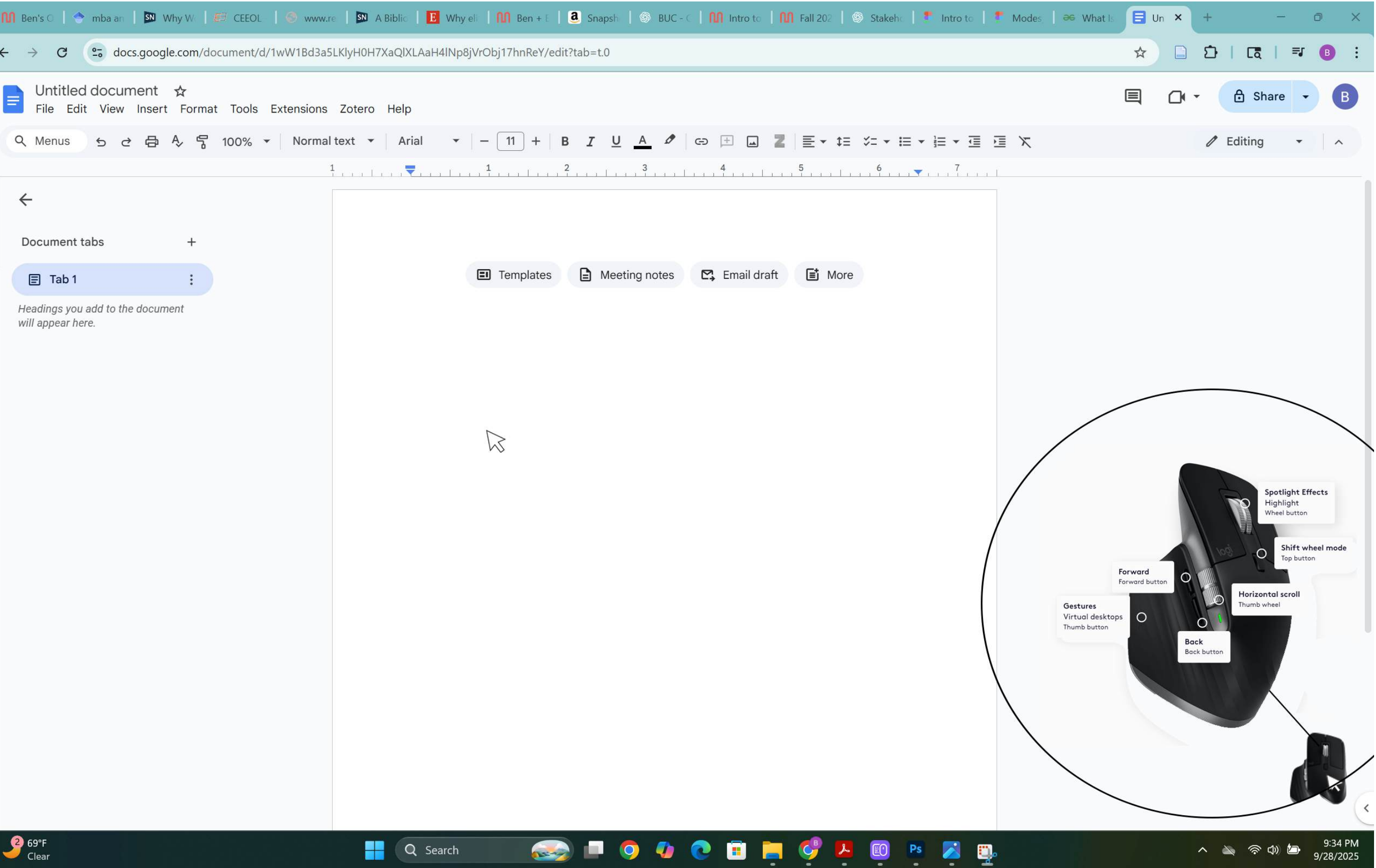
Inclusion Principle #3

Have reminders available for what you programmed each button for, helping move actions to System One

Currently, going in the app is necessary to view buttons



Final Proposition:



Concept #3: Physical Experience

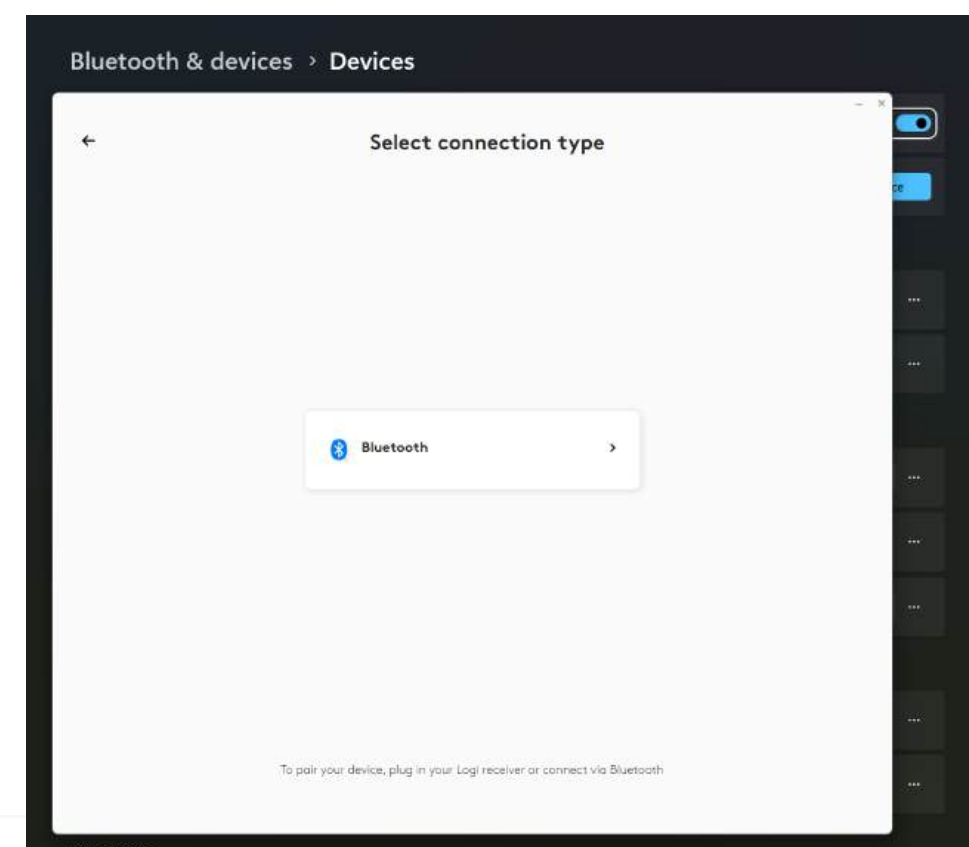
Multi-sensorial feedback

Inclusion Principle #4

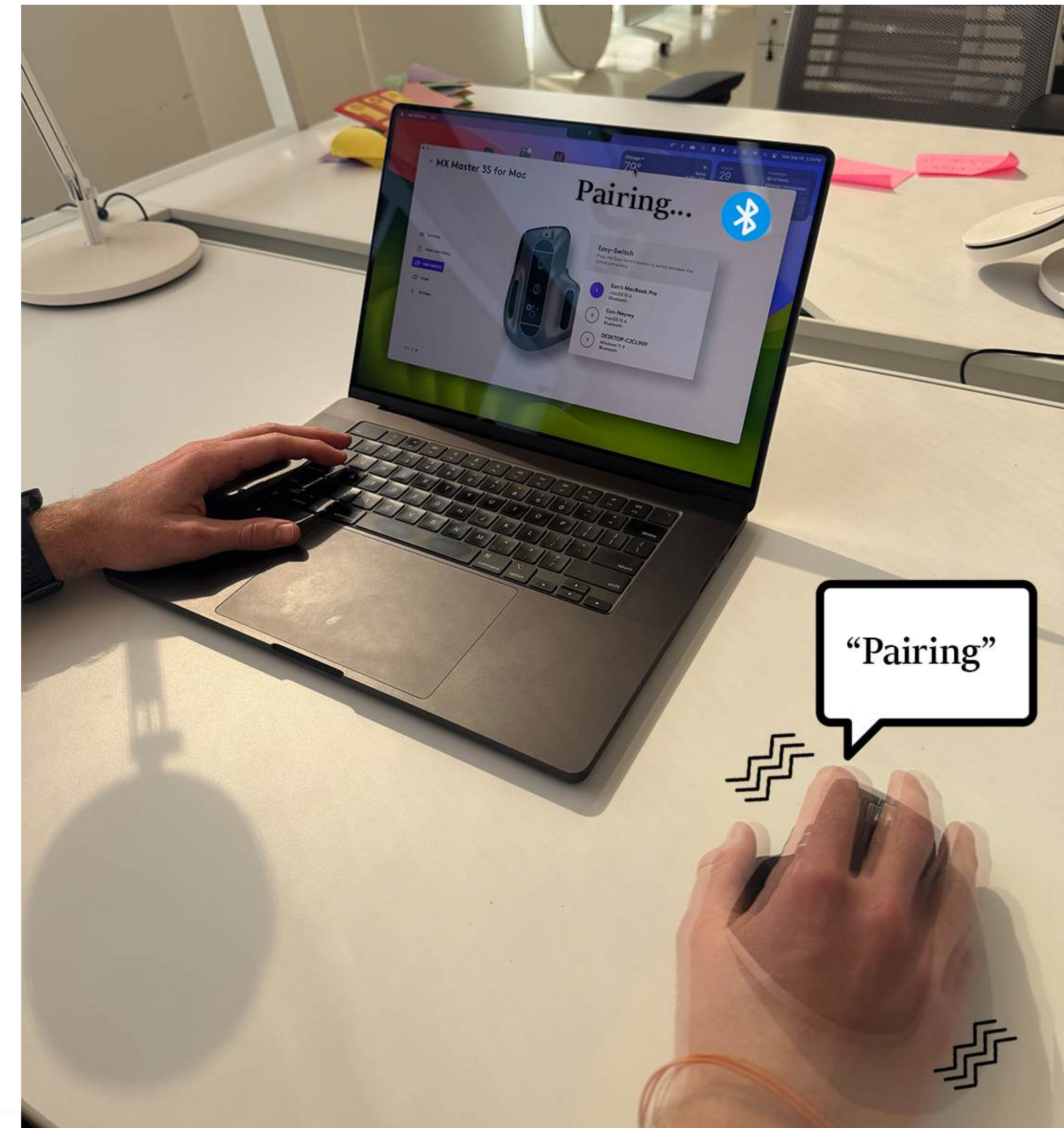
Incorporate different mechanisms of feedback to go with different cases!

- Vibration = connected successfully
- Sound = confirm connected device
- Visual = action performed

Currently, there is no indication within the app or mouse for the connection process



Final Proposition:



Thank you!

Appendix

Concept sketches

